

14. Test Closure Report

14.1 Introduction

The Test Closure Report is the final phase of the load testing lifecycle. It provides a summary of all testing activities performed, documents the overall outcomes, evaluates whether the testing objectives were achieved, and formally concludes the project.

This report serves as evidence that the load testing activities for the BlazeDemo Flight Booking Application were completed successfully and that all planned test scenarios were executed according to the project requirements.

The load testing activities were performed using Gatling, and the generated reports were analyzed to evaluate application performance under simulated user load conditions.

14.2 Project Completion Summary

The objective of this project was to perform load testing on the BlazeDemo Flight Booking Application and analyze its performance under concurrent user activity.

The project included:

- Test Planning
- Environment Setup
- Scenario Design
- Gatling Script Development
- Test Execution
- Report Generation
- Performance Analysis
- Result Documentation

All planned activities were completed successfully.

14.3 Test Objectives Achievement Status

The following table summarizes the achievement status of the project objectives.

Objective	Status
Configure Gatling Environment	Completed
Develop Load Testing Scripts	Completed
Implement Search Flight Scenario	Completed
Implement Choose Flight Scenario	Completed
Implement Complete Booking Scenario	Completed
Execute Load Tests	Completed
Generate Gatling Reports	Completed
Analyze Performance Metrics	Completed
Prepare Test Documentation	Completed

Conclusion

All project objectives were successfully achieved.

14.4 Scenario Execution Status

The following scenarios were executed during the testing process.

Scenario	Status
Scenario 1 – Search Flights	PASS
Scenario 2 – Choose Flight	PASS
Scenario 3 – Complete Booking	PASS

Observation

All scenarios completed successfully and no execution failures were encountered.

The application processed all requests and returned valid responses throughout the testing process.

14.5 Performance Testing Outcome

The load testing results demonstrated that the BlazeDemo application was capable of handling the configured workload effectively.

Key Results

- 20 Virtual Users Simulated
- 30-Second Ramp-Up Duration
- 100% Request Success Rate
- Zero Request Failures
- Stable Throughput
- Acceptable Response Times
- Successful Completion of All Business Workflows

The performance metrics collected during execution indicated that the application remained stable under the applied load conditions.

14.6 Issues Encountered During Testing

During the execution of the project, no critical issues or application failures were observed.

Observations

- No server-side errors occurred.
- No request validation failures occurred.
- No workflow interruptions were detected.
- No application crashes were observed.

Minor increases in response times were observed during the Complete Booking workflow due to additional transaction processing activities; however, these did not impact the successful completion of requests.

14.7 Risks and Limitations

Although the testing activities were completed successfully, certain limitations should be considered.

Limitations

Limited User Load

The application was tested using 20 virtual users.

Higher workloads such as:

- 50 Users
- 100 Users
- 200 Users

were not included within the scope of this project.

External Hosting Environment

BlazeDemo is a publicly hosted demonstration application.

Response times may be influenced by:

- Internet connectivity
- Network latency
- External server performance

which are outside the control of the testing environment.

Functional Scope

Only the primary flight booking workflows were included in the testing scope.

Additional application features were not considered during this project.

14.8 Deliverables Produced

The following project deliverables were successfully generated.

Gatling Simulation Files

- BlazeDemoLoadTest.java
- SearchFlightSimulation.java
- ChooseFlightSimulation.java
- CompleteBookingSimulation.java
- GatlingRunner.java

Configuration Files

- pom.xml
- gatling.conf

Generated Reports

- BlazeDemo_Load_Test_Report.pdf
- searchflightsimulation.pdf
- ChooseFlightSimulation.pdf
- CompleteBookingSimulation.pdf

Documentation

- Complete Load Testing Project Report
- Performance Analysis Documentation

- Test Closure Report
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14.9 Lessons Learned

The project provided practical experience in performance testing and load testing concepts.

Key learnings include:

- Understanding Gatling Architecture
- Creating Gatling Simulation Scripts
- Capturing HTTP Requests
- Implementing User Workflows
- Simulating Virtual Users
- Analyzing Response Times
- Evaluating Throughput Metrics
- Generating Performance Reports
- Identifying Performance Bottlenecks
- Preparing Performance Test Documentation

These learnings contribute to a better understanding of real-world performance testing practices.

14.10 Final Conclusion

The load testing project for the BlazeDemo Flight Booking Application was completed successfully using Gatling.

All planned scenarios, including Search Flights, Choose Flight, and Complete Booking, were executed successfully under a load of 20 virtual users with a 30-second ramp-up period.

The application demonstrated:

- Stable performance
- 100% request success rate
- Zero transaction failures
- Acceptable response times
- Reliable processing of concurrent user requests

Based on the observed results, the BlazeDemo Flight Booking Application successfully handled the configured workload and maintained consistent performance throughout testing.

The generated Gatling reports, performance analysis results, and execution evidence confirm the successful completion of the project and fulfillment of all stated objectives.

This concludes the Load Testing of BlazeDemo Flight Booking Application Using Gatling project.